

Graph NP-Completeness Proofs

Independent Set, Coloring, Partition, and Isomorphism Notes

Important Status Note

Graph Isomorphism Is Different

Ordinary GRAPH-ISOMORPHISM is **not known** to be NP-complete. It is in NP, but no polynomial-time algorithm is known and no NP-completeness proof is known.

If an exam asks for an NP-complete isomorphism-type problem, it usually means SUBGRAPH-ISOMORPHISM, not ordinary GRAPH-ISOMORPHISM.

Reduction Map

$$\text{CLIQUE} \leq_p \text{INDEPENDENT-SET}, \quad 3\text{-COLOR} \leq_p \text{VERTEX-COLOR}, \\ \text{VERTEX-COLOR} \leq_p \text{CLIQUE-PARTITION}.$$

Also:

$$3\text{-CNF-SAT} \leq_p 3\text{-COLOR}, \quad \text{CLIQUE} \leq_p \text{SUBGRAPH-ISOMORPHISM}.$$

For edge coloring, the standard theorem is Holyer's theorem: deciding whether a cubic graph is 3-edge-colorable is NP-complete.

1 Independent Set is NP-Complete

Problem

$$\text{INDEPENDENT-SET} = \{\langle G, k \rangle \mid G \text{ has an independent set of size at least } k\}.$$

A set S is independent if no two vertices in S are adjacent.

Reduction Used

$$\text{CLIQUE} \leq_p \text{INDEPENDENT-SET}.$$

Use the complement graph.

Proof

First, INDEPENDENT-SET is in NP. A certificate is a set S of k vertices. The verifier checks that no pair of vertices in S is joined by an edge.

For NP-hardness, reduce from CLIQUE. Given a graph $G = (V, E)$ and integer k , construct the complement graph $\overline{G}$:

$$(u, v) \in E(\overline{G}) \iff u \neq v \text{ and } (u, v) \notin E(G).$$

Output $\langle \overline{G}, k \rangle$.

A set S is a clique in G iff every pair in S is an edge of G , which is equivalent to no pair in S being an edge of $\overline{G}$. Thus

$$\langle G, k \rangle \in \text{CLIQUE} \iff \langle \overline{G}, k \rangle \in \text{INDEPENDENT-SET}.$$

The complement graph is computable in polynomial time. Since CLIQUE is **NP**-complete and INDEPENDENT-SET is in **NP**, INDEPENDENT-SET is **NP**-complete.

2 Vertex Coloring is NP-Complete

Problem

$\text{VERTEX-COLOR} = \{\langle G, k \rangle \mid G \text{ can be properly colored with at most } k \text{ colors}\}.$

A proper coloring gives different colors to adjacent vertices.

Core Fact

3-COLOR is **NP**-complete. Since 3-COLOR is just the special case $k = 3$ of VERTEX-COLOR, the general VERTEX-COLOR problem is also **NP**-complete.

3-COLOR Is in NP

A certificate is a color for every vertex. The verifier checks every edge (u, v) and confirms that u and v have different colors. This is polynomial time.

NP-Hardness Idea for 3-COLOR

Reduce

$$3\text{-CNF-SAT} \leq_p 3\text{-COLOR}.$$

Given a 3-CNF formula, build a graph with three special vertices

$$T, \quad F, \quad B$$

forming a triangle. Their three colors represent true, false, and base.

For each variable x_i , create two vertices x_i and $\neg x_i$. Connect them to each other and to B . Therefore x_i and $\neg x_i$ must receive the two colors T and F , in opposite order.

For each clause $(\ell_1 \vee \ell_2 \vee \ell_3)$, attach a constant-size clause gadget to the three literal vertices. The standard clause gadget has this property:

it is 3-colorable exactly when at least one of ℓ_1, ℓ_2, ℓ_3 has color T .

Equivalently, it cannot be colored when all three literals have color F .

Thus a satisfying assignment of the formula gives a 3-coloring of the graph. Conversely, any 3-coloring assigns each variable a truth value, and every clause gadget forces at least one literal in the clause to be true. Hence the formula is satisfiable.

The graph has polynomial size because each variable and each clause contributes only a constant-size gadget. Therefore 3-COLOR is **NP**-complete, and so VERTEX-COLOR is **NP**-complete.

Exam Note

If your instructor requires the full gadget, draw the standard 3-SAT to 3-coloring clause gadget and state its truth table. The proof above is the short exam version: palette triangle, variable gadgets, clause gadgets.

3 Graph Partition is NP-Complete

Definition Used Here

There are several problems called “graph partition.” These notes use the standard **clique partition** version: partition the vertex set into at most k cliques. This problem is also called the vertex clique-cover problem.

Problem

$\text{CLIQUE-PARTITION} = \{\langle G, k \rangle \mid V(G) \text{ can be partitioned into at most } k \text{ cliques}\}.$

Reduction Used

$\text{VERTEX-COLOR} \leq_p \text{CLIQUE-PARTITION}.$

Color classes in G are independent sets; in the complement graph they become cliques.

Proof

First, CLIQUE-PARTITION is in **NP**. A certificate is a partition of $V(G)$ into at most k parts. The verifier checks that every part is a clique.

For NP-hardness, reduce from VERTEX-COLOR . Given $\langle G, k \rangle$, construct the complement graph $\overline{G}$ and output

$$\langle \overline{G}, k \rangle.$$

If G has a proper k -coloring, then each color class is an independent set in G . Therefore each color class is a clique in $\overline{G}$, giving a partition of $V(\overline{G})$ into at most k cliques.

Conversely, if $\overline{G}$ can be partitioned into at most k cliques, then each clique is an independent set in G . Color all vertices in the same part with the same color. This gives a proper k -coloring of G . Thus

$$\langle G, k \rangle \in \text{VERTEX-COLOR} \iff \langle \overline{G}, k \rangle \in \text{CLIQUE-PARTITION}.$$

The construction is polynomial. Since VERTEX-COLOR is **NP**-complete and CLIQUE-PARTITION is in **NP**, CLIQUE-PARTITION is **NP**-complete.

4 Edge Coloring is NP-Complete

Problem

$\text{EDGE-COLOR} = \{\langle G, k \rangle \mid E(G) \text{ can be properly colored with at most } k \text{ colors}\}.$

A proper edge coloring gives different colors to edges sharing an endpoint.

Known Hard Core

Holyer’s theorem says that deciding whether a cubic graph is 3-edge-colorable is **NP**-complete. Therefore the general EDGE-COLOR problem is **NP**-complete.

Proof Using Holyer’s Theorem

First, EDGE-COLOR is in **NP**. A certificate assigns a color to every edge. The verifier checks each pair of incident edges and confirms that their colors are different.

For NP-hardness, use the known theorem that

$$\{\langle G \rangle \mid G \text{ is cubic and 3-edge-colorable}\}$$

is **NP**-complete. Given such a cubic graph G , output the EDGE-COLOR instance

$$\langle G, 3 \rangle.$$

This is clearly polynomial, and

$$G \text{ is 3-edge-colorable} \iff \langle G, 3 \rangle \in \text{EDGE-COLOR}.$$

Thus EDGE-COLOR is **NP**-hard. Since it is also in **NP**, it is **NP**-complete.

Exam Note

Unlike independent set or vertex coloring, the full NP-hardness proof for edge coloring is gadget-heavy. Unless your course proves Holyer's theorem, cite it as the known NP-complete source problem.

5 Graph Isomorphism: In NP, Not Known NP-Complete

Problem

$$\text{GRAPH-ISOMORPHISM} = \{\langle G_1, G_2 \rangle \mid G_1 \text{ and } G_2 \text{ are isomorphic}\}.$$

Two graphs are isomorphic if there is a bijection between their vertex sets that preserves adjacency and nonadjacency.

GRAPH-ISOMORPHISM Is in NP

A certificate is a bijection

$$f : V(G_1) \rightarrow V(G_2).$$

The verifier checks that f is a bijection and that for every pair of vertices $u, v \in V(G_1)$,

$$(u, v) \in E(G_1) \iff (f(u), f(v)) \in E(G_2).$$

There are only $\mathcal{O}(|V|^2)$ pairs to check, so verification is polynomial. Thus

$$\text{GRAPH-ISOMORPHISM} \in \text{NP}.$$

Do Not Write This as NP-Complete

No NP-completeness proof is known for ordinary GRAPH-ISOMORPHISM. It is one of the classic problems in **NP** whose exact status is unresolved.

For exams, write:

$\text{GRAPH-ISOMORPHISM} \in \text{NP}$, but GRAPH-ISOMORPHISM is not known to be **NP**-complete.

6 Subgraph Isomorphism is NP-Complete

Nearby NP-Complete Problem

$\text{SUBGRAPH-ISOMORPHISM} = \{\langle G_1, G_2 \rangle \mid G_1 \text{ is isomorphic to a subgraph of } G_2\}.$

Proof

$\text{SUBGRAPH-ISOMORPHISM}$ is in **NP** because a certificate is an injective map $f : V(G_1) \rightarrow V(G_2)$; the verifier checks that every edge of G_1 maps to an edge of G_2 .

For NP-hardness, reduce from CLIQUE . Given $\langle G, k \rangle$, set

$$G_1 = K_k \quad \text{and} \quad G_2 = G.$$

Then K_k is isomorphic to a subgraph of G iff G contains a clique of size k . Thus

$$\langle G, k \rangle \in \text{CLIQUE} \iff \langle K_k, G \rangle \in \text{SUBGRAPH-ISOMORPHISM}.$$

The construction is polynomial. Since CLIQUE is **NP**-complete and $\text{SUBGRAPH-ISOMORPHISM}$ is in **NP**, $\text{SUBGRAPH-ISOMORPHISM}$ is **NP**-complete.

Final Summary

Exam-Safe Status Table

Problem	Status and proof idea
INDEPENDENT-SET	NP -complete by complement reduction from CLIQUE .
VERTEX-COLOR	NP -complete because 3-COLOR is NP -complete via 3-SAT gadgets.
CLIQUE-PARTITION	NP -complete by complement reduction from VERTEX-COLOR.
EDGE-COLOR	NP -complete by Holyer's theorem for cubic 3-edge-coloring.
GRAPH-ISOMORPHISM	In NP , but not known to be NP -complete.
SUBGRAPH-ISOMORPHISM	NP -complete by reducing CLIQUE with $G_1 = K_k$.